

signal processors, and field programmable gate arrays (FPGAs).

IMUs, the onboard navigation core, use MEMS gyroscopes and accelerometers to measure angular velocities and linear accelerations through semiconductor micromachining. Their signal chains require low-noise amplification, 18-24-bit ADCs, and digital filtering to handle faint signals. Radiation-induced upsets in analogue circuits and extreme temperatures remain persistent challenges, mitigated through thermal compensation and radiation-tolerant microcontrollers.

Star trackers, specialised cameras for attitude determination, employ radiation-tolerant CCD or CMOS sensors with pixel-level redundancy. Optics focus starlight onto the array; charges are amplified on-chip, digitised, and processed via centroiding to determine sub-pixel star positions, followed by loss-in-space or catalogue-matching algorithms for absolute attitude determination. Modern radiation-hardened (system-on-chips) SoCs now integrate ARM or RISC-V cores with image-processing acceleration, replacing dedicated digital signal processors.

The problem statement: Why autonomy is non-negotiable

Spacecraft encounter a unique mix of risks: radiation-induced bit flips and latch-ups, thermal transients that alter sensor biases, micro-meteoroid impacts, intermittent communications, and unavoidable light-time delays that make real-time human correction impossible for distant missions.

For low-Earth orbit constellations, operational scale makes ground-in-the-loop micro-management impractical. For deep-space missions, hours or days of round-trip latency demand local decision-making. Navigation failures cascade: an erroneous attitude estimate degrades star-tracker performance, thereby altering thruster firings and trajectory predictions. Left

unchecked, small faults escalate into mission-threatening anomalies.

Autonomy must therefore deliver two linked capabilities: first, the ability to detect, isolate, and recover from faults locally through fault detection, isolation, and recovery (FDIR); second, the ability to maintain accurate navigation by fusing heterogeneous sensor inputs even when some sources degrade or fail. Both must operate within tight power, mass, and computational budgets while remaining robust to radiation and other adversities.

The electronics foundation: Making on-board autonomy practical

At the hardware layer, autonomous navigation and onboard FDIR require radiation-hardened computational elements, reconfigurable logic, high-fidelity sensor front-ends, and resilient data buses.

The core typically pairs a radiation-tolerant microprocessor or multi-core SoC with FPGAs. These enable hardware acceleration for real-time tasks such as filtering, pulsar pulse detection, and deep-learning inference, while also supporting reconfigurable fault-tolerant logic such as triple modular redundancy.

Analogue front-ends remain critical for attitude sensors, star trackers, sun sensors, gyroscopes, cameras, lidar, and radar, demanding low-noise ADCs, temperature compensation, shielding, and conservative margins for sample-and-hold circuits and voltage references. Actuators use PWM drivers, power telemetry, and watchdog-protected control loops with hardware interlocks.

Data buses rely on redundant space-grade controllers over fault-tolerant links such as SpaceWire or redundant CAN, with distributed voting, health telemetry, and cross-strapping for failover. Together, these elements form an architecture that supports hardware-speed detect-isolate-reconfigure loops backed

by rich telemetry for higher-level decision-making.

Autonomous fault diagnosis: Concepts and algorithms

Autonomous navigation fault diagnosis involves three tasks: detection, isolation, and recovery. Traditional aerospace FDIR relied on model-based thresholds and rule sets, such as flagging one sensor when it diverges significantly from another. Although interpretable, these approaches remain fragile against novel or correlated faults.

Modern systems combine model-based and data-driven methods. Model-based observers generate residuals by comparing predicted outputs from dynamics models or Kalman estimators with actual sensor data; significant deviations trigger fault hypotheses. Isolation uses statistical tests, including sequential probability ratio or generalised likelihood ratio methods, or banked observers for probabilistic reasoning.

Machine learning enhances the detection of complex signatures. Supervised classifiers trained on simulated data flag anomalies, while unsupervised autoencoders model healthy behaviour to identify deviations. For spaceflight, these models remain compact, explainable where possible, and validated alongside classical approaches in a layered hybrid defence that balances safety with flexibility.

Hardware and software responsibilities are divided deliberately. FPGAs handle time-critical residual checks for low latency and deterministic behaviour, while software layers manage fusion and ML inference. Redundant paths and voting schemes minimise false positives and enable graded responses that avoid unnecessary resets.

Adaptive multi-source fusion architecture

We recognise that no single navigation sensor provides complete, fault-free information across all scenarios.